

More randomized experiments

STSCI/INFO/ILRST 3900: Causal Inference

3 Sep 2026

Learning goals for today

At the end of class, you will be able to:

1. Explain conditionally randomized experiments
2. Identify the “idealized experiment” as a goal

Logistics

- ▶ After class, read Ch 2.1 and 2.2 in Hernan and Robins 2023
- ▶ Quiz 1 in class on Sep 17
- ▶ Labor Day

Exchangeability

Definition: Exchangeability means that the potential outcomes, $(Y_i^{a=1}, Y_i^{a=0})$, are independent of the observed treatment (A_i)

In mathematical notation,

$$\underbrace{Y_i^{a=1}, Y_i^{a=0}}_{\text{potential outcomes}} \perp\!\!\!\perp \underbrace{A_i}_{\text{observed treatment}}$$

Exchangeability

For outcomes: No Disease = 1, Disease = 0,

		$\gamma^{Delicious}$	$\gamma^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No disease	No disease		
	Ind 2	No disease	No disease		
	Ind 3	No disease	No disease		
	Ind 4	No disease	No disease		
Some Risk	Ind 5	Disease	No disease		
	Ind 6	Disease	No disease		
	Ind 7	Disease	No disease		
	Ind 8	Disease	No disease		

Exchangeability

For outcomes: No Disease = 1, Disease = 0,

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No disease	No disease		
	Ind 2	No disease	No disease		
	Ind 3	No disease	No disease		
	Ind 4	No disease	No disease		
Some Risk	Ind 5	Disease	No disease		
	Ind 6	Disease	No disease		
	Ind 7	Disease	No disease		
	Ind 8	Disease	No disease		

$$ATE = E(Y^{Healthy}) - E(Y^{Delicious}) = 1 - 1/2 = 1/2$$

Exchangeability

When you picked a diet knowing your potential outcomes

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Delicious	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Delicious	No Disease
Some Risk	Ind 5	Disease	No Disease	Healthy	No Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Healthy	No Disease
	Ind 8	Disease	No Disease	Delicious	Disease

Exchangeability

When you picked a diet knowing your potential outcomes

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Delicious	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Delicious	No Disease
Some Risk	Ind 5	Disease	No Disease	Healthy	No Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Healthy	No Disease
	Ind 8	Disease	No Disease	Delicious	Disease

Exchangeability

When you picked a diet knowing your potential outcomes

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Delicious	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Delicious	No Disease
Some Risk	Ind 5	Disease	No Disease	Healthy	No Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Healthy	No Disease
	Ind 8	Disease	No Disease	Delicious	Disease

$$E(Y^{Delicious} \mid A = \text{Healthy}) = 0$$

$$E(Y^{Healthy} \mid A = \text{Healthy}) = 1$$

Exchangeability

When you picked a diet knowing your potential outcomes

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Delicious	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Delicious	No Disease
Some Risk	Ind 5	Disease	No Disease	Healthy	No Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Healthy	No Disease
	Ind 8	Disease	No Disease	Delicious	Disease

$$E(Y^{Delicious} \mid A = \text{Healthy}) = 0 \quad E(Y^{Delicious} \mid A = \text{Delicious}) = 4/5$$

$$E(Y^{Healthy} \mid A = \text{Healthy}) = 1 \quad E(Y^{Healthy} \mid A = \text{Delicious}) = 1$$

Exchangeability

When you picked a diet knowing your potential outcomes

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Delicious	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Delicious	No Disease
Some Risk	Ind 5	Disease	No Disease	Healthy	No Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Healthy	No Disease
	Ind 8	Disease	No Disease	Delicious	Disease

$$E(Y^{Delicious} \mid A = \text{Healthy}) = 0 \quad E(Y^{Delicious} \mid A = \text{Delicious}) = 4/5$$

$$E(Y^{Healthy} \mid A = \text{Healthy}) = 1 \quad E(Y^{Healthy} \mid A = \text{Delicious}) = 1$$

$$E(Y \mid A = \text{Healthy}) - E(Y \mid A = \text{Delicious}) = 1 - 4/5 = 1/5$$

Exchangeability

When diet was randomly assigned:

		$\gamma^{Delicious}$	$\gamma^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Healthy	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Healthy	No Disease
Some Risk	Ind 5	Disease	No Disease	Delicious	Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Delicious	Disease
	Ind 8	Disease	No Disease	Healthy	No Disease

Exchangeability

When diet was randomly assigned:

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Healthy	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Healthy	No Disease
Some Risk	Ind 5	Disease	No Disease	Delicious	Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Delicious	Disease
	Ind 8	Disease	No Disease	Healthy	No Disease

$$E(Y^{Delicious} \mid A = \text{Healthy}) = 2/4 \quad E(Y^{Delicious} \mid A = \text{Delicious}) = 2/4$$

$$E(Y^{Healthy} \mid A = \text{Healthy}) = 1 \quad E(Y^{Healthy} \mid A = \text{Delicious}) = 1$$

Exchangeability

When diet was randomly assigned:

		$Y^{Delicious}$	$Y^{Healthy}$	A_i	Y_i
No Risk	Ind 1	No Disease	No Disease	Delicious	No Disease
	Ind 2	No Disease	No Disease	Healthy	No Disease
	Ind 3	No Disease	No Disease	Delicious	No Disease
	Ind 4	No Disease	No Disease	Healthy	No Disease
Some Risk	Ind 5	Disease	No Disease	Delicious	Disease
	Ind 6	Disease	No Disease	Healthy	No Disease
	Ind 7	Disease	No Disease	Delicious	Disease
	Ind 8	Disease	No Disease	Healthy	No Disease

$$E(Y^{Delicious} \mid A = \text{Healthy}) = 2/4 \quad E(Y^{Delicious} \mid A = \text{Delicious}) = 2/4$$

$$E(Y^{Healthy} \mid A = \text{Healthy}) = 1 \quad E(Y^{Healthy} \mid A = \text{Delicious}) = 1$$

$$E(Y \mid A = \text{Healthy}) - E(Y \mid A = \text{Delicious}) = 1 - 1/2 = 1/2$$

Why is exchangeability good?

When exchangeability is true, it implies

$$\underbrace{E(Y^{a=1} \mid A = 1)}_{\text{Within treated}} = \underbrace{E(Y^{a=1} \mid A = 0)}_{\text{Within not treated}} = \underbrace{E(Y^{a=1})}_{\text{everyone}}$$

Why is exchangeability good?

When exchangeability is true, it implies

$$\underbrace{E(Y^{a=1} \mid A = 1)}_{\text{Within treated}} = \underbrace{E(Y^{a=1} \mid A = 0)}_{\text{Within not treated}} = \underbrace{E(Y^{a=1})}_{\text{everyone}}$$

This allows us to identify the average treatment effect (ATE)

$$\text{ATE} = \underbrace{E(Y^{a=1})}_{\text{if everyone is treated}} - \underbrace{E(Y^{a=0})}_{\text{if no-one is treated}}$$

because we can plug-in

$$\underbrace{E(Y \mid A = 1)}_{\text{observed outcomes for **actually** treated}} = \underbrace{E(Y^{a=1} \mid A = 1)}_{\text{pot. outcome under treatment for **actually** treated}} = \underbrace{E(Y^{a=1})}_{\text{pot. outcome under treatment for everyone}}$$

When does exchangeability hold?

- ▶ Data does not tell us directly whether exchangeability holds
- ▶ We must know how the data was gathered
- ▶ Exchangeability holds by design in experiments

Exercise

- ▶ Exchangeability implies that $Y^a \perp\!\!\!\perp A$ for all treatment values a
- ▶ How is this different than $Y \perp\!\!\!\perp A$?
- ▶ In randomized experiments, $Y^a \perp\!\!\!\perp A$ is usually true. Is $Y \perp\!\!\!\perp A$ ever true?

Why are experiments good?

- ▶ Experiments are great because exchangeability holds by design
- ▶ To estimate the causal effect from experimental data, we can simply take the difference in observed means

Why are experiments good?

- ▶ Experiments are great because exchangeability holds by design
- ▶ To estimate the causal effect from experimental data, we can simply take the difference in observed means
- ▶ But they are also great for other reasons

Precise questions

Experiments allow us to answer precise questions

What is the causal effect of vaccination on covid?

Precise questions

Experiments allow us to answer precise questions

What is the causal effect of Two shots of the Pfizer vaccine 21 days apart on Covid?

Precise questions

Experiments allow us to answer precise questions

What is the causal effect of **Two shots of the Pfizer vaccine 21 days apart** on a **positive Covid test within 14 weeks of vaccination in 2020**?

Precise questions

Experiments allow us to answer precise questions

What is the causal effect of **Two shots of the Pfizer vaccine 21 days apart** on a **positive Covid test within 14 weeks of vaccination in 2020**?

- ▶ Experiments allow us to (more easily) specify precise treatments and outcomes

Limits of experiments

Experiments may not be possible because of

- ▶ **Feasibility:** What is the causal effect on global average temperature of decreasing global CO_2 levels by 100 ppm?

Limits of experiments

Experiments may not be possible because of

- ▶ **Feasibility:** What is the causal effect on global average temperature of decreasing global CO_2 levels by 100 ppm?
- ▶ **Cost:** What is the causal effect of giving every child 1 million USD on educational outcomes?

Limits of experiments

Experiments may not be possible because of

- ▶ **Feasibility:** What is the causal effect on global average temperature of decreasing global CO_2 levels by 100 ppm?
- ▶ **Cost:** What is the causal effect of giving every child 1 million USD on educational outcomes?
- ▶ **Ethics:** What is the causal effect on cancer of smoking cigarettes?

Making decisions with data

- ▶ Randomized experiments are powerful tools for learning causal relationships
- ▶ Experiments may have negative effect on participants or larger population¹
- ▶ Congress commissioned the Belmont Report² in 1974 in response to Tuskegee syphilis study

¹Mcdermott and Hatemi PNAS 2020

<https://www.pnas.org/doi/10.1073/pnas.2012021117>

²<https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html>

Belmont Report

- ▶ Multidisciplinary federal commission including doctors, scientists, philosophers, lawyers, civil rights leaders
- ▶ Guiding principles
 - ▶ Respect for persons: protect personal autonomy
 - ▶ Beneficence: Do no harm
 - ▶ Justice: distribute the burden/benefits fairly
- ▶ Application of principles
 - ▶ Informed Consent
 - ▶ Careful assessment of risk/benefits
 - ▶ Fair selection of subjects

Belmont Report

- ▶ Multidisciplinary federal commission including doctors, scientists, philosophers, lawyers, civil rights leaders
- ▶ Guiding principles
 - ▶ Respect for persons: protect personal autonomy
 - ▶ Beneficence: Do no harm
 - ▶ Justice: distribute the burden/benefits fairly
- ▶ Application of principles
 - ▶ Informed Consent
 - ▶ Careful assessment of risk/benefits
 - ▶ Fair selection of subjects
- ▶ Human dignity takes priority over the production of knowledge
- ▶ Causal inference (at it's best) tells you what could be, not what ought to be

Idealized experiment as goal

- ▶ Causal inference with observational data is even more important

Idealized experiment as goal

- ▶ Causal inference with observational data is even more important
- ▶ Formulate a precise causal question
 - ▶ Treatment
 - ▶ Outcome and timeframe
 - ▶ Population of interest
- ▶ Experiments are “gold standard” for estimating causal effects
- ▶ Imagine the “ideal experiment” to answer

Idealized experiment as goal

- ▶ Causal inference with observational data is even more important
- ▶ Formulate a precise causal question
 - ▶ Treatment
 - ▶ Outcome and timeframe
 - ▶ Population of interest
- ▶ Experiments are “gold standard” for estimating causal effects
- ▶ Imagine the “ideal experiment” to answer
- ▶ Try to replicate the “ideal experiment” with an observational analysis

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ “With the use of an interactive Web-based system, participants in the trial were randomly assigned in a 1:1 ratio to receive 30 μg of BNT162b2 (0.3 ml volume per dose) or saline placebo.”

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ “With the use of an interactive Web-based system, participants in the trial were randomly assigned in a 1:1 ratio to receive 30 μg of BNT162b2 (0.3 ml volume per dose) or saline placebo.”
- ▶ Suppose the researchers instead split the group into two groups
 - ▶ Age ≥ 55 : Vaccine with probability $2/3$
 - ▶ Age < 55 : Vaccine with probability $1/2$
- ▶ Does exchangeability still hold?

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ Age ≥ 55 more likely to get vaccine; more likely to get COVID even if treated
- ▶ Age ≥ 55 less likely to get vaccine; less likely to get COVID even if treated

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ Age ≥ 55 more likely to get vaccine; more likely to get COVID even if treated
- ▶ Age ≥ 55 less likely to get vaccine; less likely to get COVID even if treated
- ▶ Vaccinated individuals have $Y^{a=1}$ that are more likely to be 1 than unvaccinated individuals
- ▶ Exchangeability does not hold in entire population
- ▶ Exchangeability holds within each sub-population
- ▶ Two separate experiments; both are exchangeable

Conditional randomization

- ▶ **Marginal exchangeability:** $Y^a \perp\!\!\!\perp A$ for all a
- ▶ **Conditional exchangeability:** $Y^a \perp\!\!\!\perp A \mid L$ for all a
The potential outcomes are independent of treatment
conditional on L

Conditional randomization

- ▶ **Marginal exchangeability:** $Y^a \perp\!\!\!\perp A$ for all a
- ▶ **Conditional exchangeability:** $Y^a \perp\!\!\!\perp A \mid L$ for all a
The potential outcomes are independent of treatment **conditional on L**
- ▶ If you tell me $A_i = 1$, I learn something about $Y_i^{a=1}, Y_i^{a=0}$
- ▶ Suppose you first tell me someone's age, I learn something about $Y_i^{a=1}, Y_i^{a=0}$. Next you tell me $A_i = 1$, I don't learn anything new about $Y_i^{a=1}, Y_i^{a=0}$ (in addition to what I previously knew)

Conditional randomization

- ▶ **Marginal exchangeability:** $Y^a \perp\!\!\!\perp A$ for all a
- ▶ **Conditional exchangeability:** $Y^a \perp\!\!\!\perp A \mid L$ for all a
The potential outcomes are independent of treatment **conditional on L**
- ▶ If you tell me $A_i = 1$, I learn something about $Y_i^{a=1}, Y_i^{a=0}$
- ▶ Suppose you first tell me someone's age, I learn something about $Y_i^{a=1}, Y_i^{a=0}$. Next you tell me $A_i = 1$, I don't learn anything new about $Y_i^{a=1}, Y_i^{a=0}$ (in addition to what I previously knew)

If $B \perp\!\!\!\perp C \mid D$

$$E(B \mid C, D) = E(B \mid D)$$

Conditional randomization

- ▶ Most useful as an idealized experiment to target with observational analysis
- ▶ Marginal exchangeability is very unlikely in observational data
- ▶ Conditional exchangeability may be more reasonable

Learning goals for today

At the end of class, you will be able to:

1. Explain conditionally randomized experiments
2. Identify the “idealized experiment” as a goal